

Dry Needling for Arthrogenic Muscle Inhibition of Quadriceps Femoris in Patients after Reconstruction of Anterior Cruciate Ligament: a Protocol for a Randomized Controlled Trial

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Background: Dry needling (DN) is recommended as a therapeutic modality for various neuromusculoskeletal disorders. No study has been performed on the impact of DN on arthrogenic muscle inhibition (AMI) after anterior cruciate ligament reconstruction (ACLR). This study protocol is aimed to investigate the impacts of DN on AMI of quadriceps femoris, corticomotor, and spinal reflex excitability in patients with ACLR.

Methods: A double-blind, between-subject, randomized, controlled trial will be conducted to measure changes in AMI after DN. Twenty-four subjects with ACLR will be recruited to receive a DN or a sham DN, providing that they met the inclusion criteria. Three sessions of DN on the quadriceps femoris will be applied during a one-week period. The primary outcome measures are the active motor threshold, motor evoked potential, and $H_{max} - M_{max}$ ratio. The secondary outcomes are the International Knee Documentation Committee subjective knee form questionnaire score and maximum quadriceps isometric torque. Data will be collected at baseline, immediately after the first session, after the third session, and at the one-month follow-up visit.

Discussion: The results of this study will provide preliminary evidence regarding the effects of DN on AMI of quadriceps femoris in patients with ACLR.

Keywords: ACL reconstruction, Arthrogenic muscle inhibition, Dry needling, Corticomotor excitability

INTRODUCTION

Rupturing the anterior cruciate ligament (ACL) is known as one of the most common knee ligament injuries [1]. Studies indicate that there are between 80,000 and 250,000 ACL injuries annually, with female athletes being between 2 and 10 times more likely to rupture their ACL than males [2]. Approximately 7% of patients suffer an ACL injury in their reconstructed limb, while a further 8% suffer an injury to their contralateral limb, even with the high success rate of surgery and physical treatment [3]. The annual cost of treatment has been estimated at USD 17.7 billion [4,5].

ACL injuries in athletes often render them being unable to participate in their sport, and in some cases, they may even lose an entire season depending on the sport. Moreover, secondary injuries may also develop, such as osteoarthritis, meniscus injury, muscle atrophy, recurrent pain, gait disorders, and knee extension defects [6,7].

Studies on muscle recovery following ACL reconstruction (ACLR) indicate that the function of the quadriceps is potentially not recovered to pre-injury levels using traditional rehabilitation techniques [8]. Therefore, understanding the causes related to the deficiencies in quadriceps strength may result in early and successful management [9]. Recent

studies suggest that after undergoing ACLR, neural changes in electrocortical brain activity, the excitability of spinal reflex, and corticospinal pathways as well as in the central activation may contribute to the inadequate improvement of quadriceps strength and function [10-12]. These neural deficits have been associated with muscle weakness in the quadriceps, disordered movement patterns, and impairments in the voluntary activation of both the reconstructed and non-reconstructed limbs [10,13,14]. This diminished ability in subjects with ACLR to fully contract the quadricep muscles is known as arthrogenic muscle inhibition (AMI). Therefore, it is important to address and manage AMI to optimize the recovery of the quadricep muscles and restore their function.

Dry needling (DN) is a relatively novel technique that is used by physiotherapists, which involves using a fine needle to puncture the skin, subcutaneous tissues, and muscles to treat a variety of neuromusculoskeletal conditions—for example: DN has been used extensively to treat musculoskeletal pain [15-17]. A meta-analysis provided evidence that DN could be used as an effective intervention for pain reduction in various musculoskeletal conditions [18]. Moreover, DN has been shown to be an effective and useful intervention for pain and function, when combined with physiotherapy [15,19]. Patients with various conditions, such as those with latent trigger points, may suffer from muscle weakness and movement dysfunction. Previous investigations have demonstrated that when combined with other interventions, DN has positive effects on muscle strength [20-22]. A recent study found that there was a significant increase in the gluteus muscle force production immediately after DN was applied to the gluteus medius [23]. A systematic review of 10 studies (n = 3,271) analyzed physiotherapy interventions that had been implemented in standard rehabilitation programs in patients after ACL reconstruction and found that most physiotherapy interventions using DN in early-stage postoperative ACL rehabilitation result in a reduction in pain and edema, while improving the range of motion (ROM), knee muscle strength, and knee function [24]. Further, a randomized, single-blinded, clinical trial, which included 44 subacute patients with ACLR also demonstrated improvements in the ROM and knee function [25]. Moreover, a study that evaluated the effects of one session of DN on EMG and passive mechanical properties of the quadriceps muscles in a group of 20 physically active male patients, who were in an advanced phase of the ACL rehabilitation program, found small improvements in the pain intensity and knee flexion ROM [26].

Knowledge of the precise mechanisms underlying the efficacy of DN remains limited. However, several mechanisms have been proposed, including an augmentation of the blood flow, which leads to enhanced tissue oxygenation, a

reduction in pain-related biochemicals, and the alleviation of contractures through the mechanical manipulation of stiff soft tissues [27-29]. It is plausible to hypothesize that localized increases in blood flow, alterations in tissue biochemistry, and improvements in cross-bridge formation may collectively contribute to the enhancements in force production and muscle strength following DN treatment. Furthermore, it is conceivable that the therapeutic benefits of DN could be associated with neurophysiological changes involving the central nervous system (CNS) [30-33]. For instance, a study involving 19 adults with chronic non-traumatic shoulder pain and infraspinatus myofascial trigger points demonstrated that DN reduced the active motor threshold and increased corticospinal excitability [34]. These findings suggest a potential association between DN and neurophysiological adaptations within the CNS that merit further investigations.

Currently, no study has assessed the effects of DN on AMI in subjects with ACLR. Therefore, the aim of this study is to examine the effects of DN on AMI in patients following an ACLR. We hypothesized that DN will provide neuromodulation effects that significantly improve the AMI and quadriceps muscle activity in subjects with ACLR.

MATERIALS AND METHODS

1. Study design

This study follows the Standard Protocol Items: Recommendations for Interventional Trials (SPIRIT) statement guidelines and is a randomized controlled trial (RCT), where both the patients and assessors will be blinded to the interventions. Approval for this study is granted by the Ethics Committee of Tehran University of Medical Sciences (TUMS), Reference ID: IR.TUMS.FNM.REC.1402.023. All individuals who agreed to participate in this study will be asked to provide their written informed consent prior to data collection.

2. Informed consent

The participants will be provided a detailed explanation of the study assessments and DN treatment. Participants will be allowed to withdraw from the study at any time without providing an explanation or fear of any consequences. The physiotherapist assessing the participants will obtain the written informed consent.

3. Study population

Eligible individuals with ACLR will be recruited for this study during the period of July 2023 and February 2024.

4. Inclusion and exclusion criteria

Inclusion criteria: 1) Patients with a history of early

unilateral ACLR that used a patellar tendon autograft at least six months prior to commencing the study; 2) the approval of a healthcare provider to resume full recreational and athletic activities; 3) aged between 18 and 40 years; 4) a normal body mass index of 18.5-24.9; 5) engaged in moderate-intensity exercise for a minimum of 30 minutes, between three to five times a week [35]; 6) no lower limb diseases or injuries in the previous two years that required surgery; 7) limb symmetry indices greater than 1.00 (the torque values of the healthy limb divided by the torque value of the reconstructed limb).

Exclusion criteria: 1) Simultaneous rupture and injury of other knee parts (e.g. menisci); 2) history of previous neurologic disorders (e.g., neuropathic pain in the lower extremities, lumbosacral radiculopathy); 3) intolerance or fear of DN, cardiopulmonary disorder, or an inability to complete 30 minutes of aerobic exercise; 4) a history of DN in the last 3 months; 5) TMS contraindications (e.g., psychiatric or epilepsy disorders, presence of metal or electronic implants, a metal foreign object in the eye, history of head trauma with loss of consciousness, pregnancy).

5. Procedure

The study will be performed at the Physiotherapy Clinic, School of Rehabilitation, TUMS. Patients will be recruited from the Physical Therapy Clinics and University Firoozgar Hospital in Tehran, Iran. The procedure and the administration of the sham DN will subsequently be described in detail. Information relating to the sham DN will be provided as a clear and concise overview detailing the necessity for implementing a sham DN in our study and its role in determining the effectiveness of the actual DN and scientific validity to ensure unbiased results. Additionally, transparency on how the sham DN is designed to have no therapeutic effect will be maintained throughout. Furthermore, as the sham DN does not provide any direct therapeutic effects, it also does not present the patients with any discomfort or cause any side effects. The participants will be informed that the study has been reviewed by the scientific review board and the Ethical Committee of TUMS to ensure the safety of the participants and that ethical guidelines are adhered to throughout. Moreover, the participants will be encouraged to ask any questions relating to the interventions, express any concerns, and will be also provided

Timepoint	Study period				
	Enrolment	Allocation	Post-allocation		
	$-t_1$	t_0	t_1	t_2	t_3
Enrolment					
Patient recruitment	x				
Eligibility screen	x				
Informed consent	x				
Allocation		x			
Interventions					
[Real dry needling]			◆————◆		
[Sham dry needling]			◆————◆		
Assessments					
[Age, sex, BMI, dominant foot]		x			
[MEPs]		x	x	x	x
[aMT]		x	x	x	x
[H_{max}/M_{max} ratio]		x	x	x	x
[IKDS score]		x	x	x	x
[MQIT]		x	x	x	x

SPIRIT = Standard Protocol Items: Recommendations for Interventional Trials; aMT = active motor threshold; MEPs = motor evoked potentials; IKDC = International Knee Documentation Committee; MQIT = maximum quadriceps isometric torque; $-t_1$ = pre-study, screening/consent; t_0 = pre-study, baseline/randomization; t_1 = immediately after the first session; t_2 = after the third session; t_3 = one month follow-up visit.

Fig. 1. SPIRIT study periods for various stages of the study.

with sufficient time to consider their participation in the study. After agreeing to participate in the study and signing the consent form, the patients will be randomly assigned to either the real DN or sham DN group. An experienced physiotherapist, who is blinded to the group randomization, will assess the patients at the baseline, immediately after the first session, after the third session, and one month after the third session. All information relating to the participants and study data will be stored on a password-secured PC so that nobody will accidentally access it. The data will be carefully entered, reviewed, filed, and stored in numerical order. To ensure the data is entered accurately, each entry will be double-checked to minimize errors. The SPIRIT study periods for various stages of the study procedure are depicted in Fig. 1.

6. Randomization and blinding

Eligible and consenting patients referred by an orthopedic surgeon will be randomly allocated to either the DN or sham DN group. To randomly assign the patients, opaque, sealed envelopes will be used to ensure the concealment of allocations, with the two blocks of 12 patients using an online research randomizer sequence generator (<http://www.randomizer.org>). The assessors and patients will be remained blind to the type of treatment being received.

7. Sample size

A scoping review examined six RCTs by comparing DN to either a sham or placebo DN, with two studies specifically focusing on the lower extremities. DN was found to provide significant improvements, with the outcomes in three out of the six trials exceeding the minimally clinically important difference [36]. Given the absence of similar studies with effective sample sizes, we determined a preliminary sample size of 24 patients as part of a pilot study, with 12 patients allocated to each group. This calculation assumed a moderate effect size of 0.5 [25,36], significance level $\alpha = 0.05$, and power of $\beta = 0.8$. The data collected during this pilot phase will be used to calculate the actual statistical power of the study. An increase in the number of subjects will be considered in the event of low statistical power to achieve these predefined levels. The orthopedic surgeons and physiotherapists will continue screening patients for eligibility until the required sample size is achieved.

We considered a strategy of short study duration with three treatment sessions, three minutes of DN, and a one-month follow-up to improve the adherence of participants to the study protocol and complete the data collection as planned. In the event of dropouts, noncompliance, or missing outcomes, an intention-to-treat analysis will be performed.

8. Interventions

The STRICTA guidelines is followed for clear and accurate reporting of DN intervention [37]. Patients in the DN and sham groups will receive DN-style treatments at the same points. The patients in the experimental group will undergo three sessions of deep DN in the quadriceps femoris muscle every other day (each point being needled for one minute), while those in the control group will receive non-penetrating sham DN. All the DN procedures will be performed by a licensed and experienced physiotherapist who possesses an MSc degree in physiotherapy and an official license to practice physiotherapy and conduct DN treatments. The physiotherapist responsible for delivering the DN has five years of experience in physiotherapy, treating people with various orthopedic conditions, including ACLR. Stainless steel, sterilized needles (DongBang AcuPrime Ltd., Korea), with a width of 0.3 mm and a length of 50 mm will be used. The procedure will follow the cone-shaped, fast-in, and fast-out technique, which involves the rapid insertion and removal of the needles. For the patients in a supine lying position with the leg in an extended neutral position, three defined points will be targeted, namely the rectus femoris (RF), vastus medialis (VM), and vastus lateralis (VL), with each point being needled for one minute, consistent with previous studies [38]. Table 1 shows the points in the quadriceps muscle where DN will be performed.

Sham DN is chosen to evaluate the placebo effects. For the sham DN procedure, a 4 cm plastic monofilament (force 10 gr, Gima Medical Co, Italy) will be used for the non-needling technique. The protocol employed for the sham DN will mirror the experimental group technique and will be administered at the same locations as the experimental group, with each point receiving sham treatment for 1 minute during each session (totaling 3 minutes) for a total of 3 sessions.

9. Outcome measures

The primary outcome measures are the active motor

Table 1. The points for dry needling of quadriceps individual muscles

Muscle	Point location
Rectus femoris	Mid-point of a line from the superior border of the patella to the ASIS
Vastus medialis	At approximately 25% of the distance between the medial superior border of the patella and the ASIS
Vastus lateralis	Mid-point of a line from the lateral superior border of the patella to the apex of the greater trochanter

ASIS = anterior superior iliac spine.

threshold, motor evoked potential, and $H_{\max} - M_{\max}$ ratio. The secondary outcomes are the International Knee Documentation Committee subjective knee form (IKDC) questionnaire score and maximum quadriceps isometric torque. All outcomes will be measured at baseline, immediately after the first session, after the third session, and at one month after the third session.

10. Assessments

1) Primary outcome measures

(1) Corticospinal excitability (active motor threshold and motor evoked potential)

Motor evoked potentials (MEPs), which are induced during transcranial magnetic stimulation (TMS), will be used to measure the corticospinal excitability. The MEP measures how much excitability can travel down the descending corticospinal tract and into the alpha motor neuron. Following routine electromyography preparation, two 10 mm pre-gelled Ag–AgCl (EL503, BIOPAC Systems Inc., Goleta, CA, USA) disk-shaped surface EMG electrodes, with an inter-electrode difference of 1.75 cm, will be placed over the distal VM muscle belly [39]. Participants sit in the testing chair during the test with their knees and hips flexed 90 degrees. The head of each participant will be covered by a lycra swim cap with a 0.5 cm grid to identify the motor cortex location and corticospinal neuron origin [40]. A double-cone-angled TMS coil (D-B80, MagVenture Inc., Atlanta, GA) will be positioned over the intersected gridlines, which will be moved in 0.5 cm increments anterior to posterior and medial to lateral until the optimal stimulating point is identified. This point is the location that produces the highest MEP amplitude in the VM [41].

The next step is to identify active motor thresholds (AMTs) by determining the lowest TMS intensity necessary to elicit a detectable (> 100 V) MEP in five out of ten trials [42]. Five MEPs will be elicited at 120% of the AMT. Following TMS testing, which employs peripheral electrical nerve stimulation, an average of the five peak-to-peak MEP amplitude data will be calculated and normalized to the average of the three maximal muscular responses elicited at rest [41]. Participants in the test produce an isometric knee extension contraction at 10% of their maximum muscle force production, which the researcher objectively assesses and monitors using a belt-stabilized handheld dynamometer (micro FET; Hoggann Scientific LLC, West Jordan UT), while also verbally providing feedback to the participant during the contraction [43].

(2) Spinal reflex excitability

Surface electromyography will be used to bilaterally collect

the quadriceps H-reflex (MP150; BIOPAC Systems, Goleta, CA). Disposable 10 mm pre-gelled silver–silver chloride electrodes will be used to apply signals to the skin that has been shaved, debrided, and cleaned with isopropyl alcohol. The gain of the electrodes is 1,000. The electrodes will be positioned 2 cm apart, parallel to the fiber orientation, and superficial to the VM muscle. A dispersive electrode will be placed on the ipsilateral posterior thigh, with the stimulating electrode placed over the femoral nerve in the inguinal fold [44]. The participants will be placed on a treatment table in the supine position, with their knees flexed to approximately 15 degrees, and instructed to relax during testing. Short duration (1-millisecond) square-wave stimulations will be triggered manually, with a minimum rest period of 10 seconds applied between stimulations, until the motor wave (M-wave) and maximal peak-to-peak amplitude of H-reflex is observed. Electromyography data will be notch-filtered at 60 Hz and bandpass-filtered between 10 and 500 Hz. The mean peak-to-peak amplitudes will be used to calculate the H-reflex to M-wave ($H_{\max}:M_{\max}$) ratio [45]. The $H_{\max}:M_{\max}$ ratio presents the percentage of the total motor neuron pool that can be recruited [46].

2) Secondary outcome measures

(1) Self-reported function

To calculate the self-reported function, each subject will complete the Persian version of the IKDC. The IKDC is a validated tool that asks patients about their symptoms, sports and daily activities, and current knee functions [47]. It consists of 18 items: 7 that relate to symptoms, 1 on their participation in sports, 9 about daily activities, and 1 on their current knee function. Response options vary for each item. Questions 1, 4, 5, 7, 8, and 9 use 5-point Likert scales, questions 2, 3, and 10 use 11-point numerical rating scales, and question 6 divided the response into a yes/no binary answer. The sum of all 18 questions produces a final IKDC score between 0 and 100 [48]. The Persian version of the IKDC has been shown to be reliable and valid for Iranian people after ACL surgery and various knee injuries [49].

(2) Maximum quadriceps isometric torque

The maximum quadriceps isometric torque will be measured using a hand-held dynamometer (HHD). To perform HHD testing, the patient is in a seated position, with their legs hanging from the edge of a treatment table, and the knee flexed at approximately 90°. A strap will be used to stabilize the thighs to the table, and a small wedge bolster will be placed at the posterior aspect of the distal thigh to reduce posterior thigh discomfort. Finally, the participants will be asked to keep their arms crossed during testing to isolate the quadriceps femoris muscle. The HHD will be fastened against

the table leg with a strap looped through a foam pad placed across the anterior shin in a location consistent with the isokinetic dynamometer (5 cm proximal to lateral malleolus). To reduce the belt slack (i.e., compliance), a small elastic wrap will be placed between the table leg and the triceps surae muscle. Three isometric contractions will be performed by participants as a warm-up at 50%, 75%, and 100% effort, with one minute of rest allowed between each contraction. Following the warm-up, the participants will complete six isometric contractions, which will be held for approximately 3 seconds, with each contraction again followed by one minute of rest. To obtain the maximum quadriceps isometric torque, the highest isometric force of the muscle will be multiplied by the torque arm (the distance between the lateral knee joint line to the distal aspect of the lateral malleolus), and 5 cm subtracted [50].

(3) Adverse events

Dry needling employs the use of thin filiform needles to puncture the muscle, meaning there is a risk of causing an adverse event (AE). The AEs may be minor (e.g., pain, bleeding, bruising) or major (e.g., pneumothorax, excessive bleeding, fainting). A study of 7,629 DN treatments by 39 physiotherapists in Ireland found that while DN resulted in some minor AEs, no major adverse events occurred [51]. A further study, which included the participation of 420 physiotherapists, was performed to determine the type of AEs associated with DN [52]. The authors found that the AEs that occurred during 20,464 DN treatment sessions were predominantly minor adverse events (i.e., mild bleeding, bruising, and pain), while major AEs were rare [52]. A recent scoping review on dosage and AEs from DN reported bleeding, bruising, and pain as common minor AEs and pneumothorax, subdural hematoma, and infection as less common major AEs [36]. The assessor will document the possible AEs caused by DN in the treatment session and at the follow-up visit.

11. Data monitoring

An independent committee comprising experts from various rehabilitation medicine disciplines will oversee the methodology to ensure that the proposed methods are adhered to and that the data is accurately collected.

12. Data collection and analysis

The data will be analyzed using the SPSS software version 24 (SPSS, Inc., Chicago, IL, USA). Descriptive statistics of frequency or mean and standard deviation will be calculated. The Kolmogorov-Smirnov (KS) test will be used to assess the data distribution. Demographic characteristics (age, height, weight, body mass index, dominant leg, and affected leg)

and baseline data will be compared using an independent t-test or Mann-Whitney U test. The Chi-square test will be used to compare the categorical variables of gender, dominant leg, and affected leg. Four measurements will be recorded for the primary and secondary outcome measures, and a mixed model, two-way repeated measure analysis of variance (ANOVA) will be used to determine the differences between groups, "Time" effects, and "Time by Group" interactions. The Bonferroni test will be used to analyze pairwise comparisons between the four tested time points. The Greenhouse-Geisser test will be used if the significance is found in Mauchly's test of variances, which indicates heterogeneity. The between-group effect sizes will be calculated using Cohen's *d*. An effect size of 0.2 is considered small, 0.5 moderate, and 0.8 large. *p*-values lower than 0.05 are considered statistically significant.

DISCUSSION

The protocol in the present study is planned to determine the effectiveness of real DN vs. sham DN in subjects after ACLR with weakness in the quadriceps femoris, subsequent to a phenomenon of AMI, and with a short-term follow-up. We hypothesized that there would be a significant difference in the efficacy of DN relative to sham DN in the treatment of AMI in subjects with ACLR. Therefore, we expect significant between-group differences in the primary outcome measures immediately after the end of treatment and at one-month follow-up, such as in the active motor threshold, motor evoked potential, and H_{max}/M_{max} ratio, and in the secondary outcomes of the IKDC score and maximum quadriceps isometric torque. To the best of our knowledge, this study is the first to analyze the effects of DN on quadriceps femoris AMI in subjects after ACLR.

The majority of subjects with weakness in the quadriceps femoris post-ACLR have been treated with time-based physiotherapy interventions, including various active and strengthening exercises. These interventions aim to improve the strength of the quadriceps femoris muscle and function. However, individuals with ACLR may still suffer from weakness in the quadriceps femoris due to the AMI. A systematic review and meta-analysis provided evidence of the deficits in corticospinal excitability and differences in neural excitability in individuals with ACLR, compared to healthy subjects (e.g., greater motor threshold, lesser MEP) [9]. Thus, the authors suggested that long-term therapeutic interventions are needed to address the corticospinal excitability and optimize the quadriceps strength [9]. Previous studies have demonstrated substantial potential benefits following DN treatment, such as an improvement in neurophysiological measures [53,54], brain activity [32,33,55], and a reduction

in the active motor threshold as well as increases in corticospinal excitability [34]. The AMI is essentially a neurological problem, whereby changes in brain activity and motor pathways result in the inhibition of the quadriceps femoris and a limitation in the ability to activate the muscle. The protocol used in our study will primarily evaluate the effects of DN on neurophysiological measures. Therefore, it appears logical to postulate that DN will cause significant effects that aid in overcoming AMI and increasing voluntary contractions in the quadriceps femoris.

This trial is limited since it does not evaluate the long-term effects of DN. We understand that a study investing the long-term outcomes would provide more clinically relevant evidence. However, there is currently no study that evaluates the use of DN in subjects with ACLR and AMI; thus, we planned to begin by investigating the short-term effects before conducting a long-term study. DN is selected for treatment as it is a relatively novel option, has immediate positive effects, improves muscle strength [56], is cost-effective [57,58], and its impacts on subjects with ACLR and AMI are unknown. The strengths of the protocol used in the present study are that it provides a high-quality design of a double-blind RCT, which evaluates the possible mechanisms for the effects of DN on AMI as well as the relevant clinical measures. The results of our study will provide evidence of the effects of DN on the AMI after ACLR and the possible mechanisms involved.

CONCLUSIONS

The treatment of AMI after ACLR is difficult and complex and may require a long-term intervention. Currently, there is no proven treatment option for AMI in subjects with ACLR. Therefore, it is imperative to consider an intervention with immediate impacts following a well-designed and feasible study protocol to optimize the strengthening of the quadriceps femoris. The use of a simple, affordable, and cost-effective intervention is important for clinicians. The results of the present trial document the effectiveness of DN and possible mechanisms in subjects with ACLR and AMI.

The results obtained in the present study will be published in a peer-reviewed journal following the performance of data collection and analysis.

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AUTHORS' CONTRIBUTIONS

All the authors were involved in the conception, design, and methodology of the protocol. MZ drafted the manuscript. NNA revised the manuscript for critically intellectual content. SH and SN reviewed and edited the manuscript. All authors read and approved the final manuscript for submission.

ETHICAL APPROVAL

Approval for this study is granted from the Ethics Committee of Tehran University of Medical Sciences (TUMS), Reference ID: IR.TUMS.FNM.REC.1402.023 which will be carried out in line with the Declaration of Helsinki.

CONSENT TO PARTICIPATE

All individuals who agree to participate in this study will be asked to give their written informed consent before data collection.

MANUSCRIPT WRITING

Artificial intelligence was not used at any stage of writing the manuscript.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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